Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*AI-DPP4Circularity: AI-Enhanced Digital Product Passport Management for the Entire Product Lifecycle in the Era of the Circular Economy*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Information & Communication		Mechanical, Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions	Clean Production	Process and Production Management Technologies
Concept of Proposed Technology	The AI-DPP4Circularity project will create a technology that uses artificial intelligence (AI) to build and manage Digital Product Passports (DPPs), which are digital records that contain key product information throughout the entire lifecycle, from design and materials to use and end-of-life. The technology will automatically extract, organize, enrich, and update product data from many different sources to create interoperable, standards-compliant DPPs that can be used by manufacturers, regulators, recyclers, and consumers. Potential uses include: • Helping companies comply with product data requirements and circular economy regulations; • Enabling better lifecycle decisions, such as repair, reuse, and recycling; • Supporting supply chain transparency and sustainability reporting; • Providing consumers with product information for informed purchasing and end-of-life			

	choices. The project will build on the FA³ST Ecosystem [1] to deliver scalable, interoperable DPPs and ensure compatibility between Korean (KRDPP) and German (DPP4.0) [2] systems for global use.
Rationale for Joint R&D Project	The Digital Product Passport (DPP) requirement under the EU's Ecodesign for Sustainable Products Regulation (ESPR) will become mandatory for priority product groups from around 2026–2030, requiring digital lifecycle data on products sold in the EU. Non-compliant products risk losing access to the EU market, which is one of South Korea's key export destinations. The South Korean DPP market is rapidly expanding, with an estimated value of USD 265 million in 2024 projected to grow to about USD 1.16 billion by 2030, reflecting strong demand for digital product data and compliance solutions. South Korea's KRDPP initiative aims to digitalize product data across lifecycle stages. The EU's DPP efforts, including standards for interoperability like the Asset Administration Shell (AAS), share similar goals of transparency and circularity but arise in a different regulatory context. Aligning KRDPP with EU DPP standards is crucial for Korean companies to ensure ongoing EU market access and to avoid fragmentation in global supply chains. Korea–Germany collaborative R&D can help co-develop interoperable DPP solutions that support compliance, reduce redundant efforts, and facilitate global integration of product passport systems.
Key Words	1. AI-Enabled DPP Generation 2. AAS-Based Interoperability 3. FA³ST Ecosystem Integration 4. Lifecycle Traceability for Circular Economy Enablement 5. Standards-Compliant Data Exchange & Regulatory Alignment

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	AI-DPP4Circularity will develop a modular AI-based multi-agent system that automatically generates, enriches, and validates AAS-compliant DPPs from heterogeneous documents, semantic ontologies, and industrial domain standards. The system aims to help SMEs create and manage DPPs efficiently and transparently by simplifying complex standards (such as AAS, DPP4.0, ESPR) through AI-enhanced workflows and automated validation. The system will use specialized agents for information retrieval, model generation, and compliance validation, employing techniques such as Retrieval-Augmented Generation (RAG) to extract relevant data and convert it into consistent, standardized models. It will process machine-specific documentation (e.g., manuals and specifications) and circularity-related data (e.g., product carbon footprint and material composition), linking extracted information to concepts in semantic background ontologies, including those developed within the CIRPASS/CIRPASS-2 ecosystem to improve precision and semantic coherence. The output will be AAS-compliant DPPs that support lifecycle transparency and traceability across design, production, use, and end-of-life contexts, aligning with emerging regulatory frameworks and circular economy goals. To evaluate system performance, generated DPPs will be benchmarked against manually created reference models, enabling quantitative assessment of completeness, accuracy, and compliance. AI-DPP4Circularity will be applied in two industrial sectors for which the DPPs will be mandatory from 2027: the textile industry and aluminum/steel.			
R&D Target Specifications	KR1: Data Extraction Accuracy (Precision ≥ 90 %, Recall ≥ 85 %) KR2: AAS/DPP Model Compliance (Schema Compliance ≥ 95 %, Semantic Consistency ≥ 95 %) KR3: Coverage of Required Attributes (≥ 95 % coverage for mandatory fields) KR4: Processing Efficiency (End-to-End DPP creation time ≤ 5 minutes per product in typical industrial use cases)			
R&D Plan	*Predevelopment phase* WP1: Predevelopment & Proof-of-Concept (M1-M6) • Task 1.1: Use case & requirements analysis (M1–M3) • Task 1.2: Domain-specific ontologies aligned with CIRPASS-II (M1–M6) • Task 1.3: DPP submodel template for circular by design products (M2–M6) • Task 1.4: Technical specifications & functional architecture (M3–M6) • Task1.5: Implement proof-of-concept demonstrator (M4 - M6) *Project phases*			

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	WP2: AI-DPP4Circularity core technologies (M7-M30) • Task 2.1: AI-assisted generation of DPPs from textual sources (M7–M18) • Task 2.2: Semantic contextualization for DPP enrichment (M13–M21) • Task 2.3: Integration with ERP/PLM/etc. systems for DPP input (M15–M24) • Task 2.4: Data extraction from product sheet diagrams for DPPs generation (M13–M21) • Task 2.5: Integration with FA³ST ecosystem for DPP use (M21–M27) • Task 2.6: Integrated testing & CI/CD (M25-M30) WP3: AI-DPP4Circularity demonstration & evaluation (M19–M36) • Task 3.1: Pilot setup & validation plan (M19–M21) • Task 3.2: Pilot 1 & Pilot 2 execution (M22–M33) • Task 3.3: Cross-case analysis & scalability (M30–M36) WP4: Impact, sustainability & standardization alignment (M1–M36) • Task 4.1: Communication & dissemination (M1–M36) • Task 4.2: Standardization & regulatory alignment (M1–M36): Engage with EU and Korea DPP standardization and regulatory initiatives to align project outcomes with evolving standards (e.g., DPP4.0, ISO 59040) and ensure interoperability and transferability. • Task 4.3: Business models and sustainability plans (M19–M36)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	FhG IOSB will provide the following contributions: in WP1, it will develop DPP submodel templates for circularity and model domain-specific ontologies for the pilots; in WP2, it will develop new methods for DPP creation based on RAG and ontology integration and integrate the results with the FA³ST Ecosystem; in WP3, it will support pilots in creating DPPs and evaluating the results; and in WP4, it will write scientific publications, participate in relevant fairs, post project results on social media, and contribute to standardization activities, including mapping between KRDPP and DPP4.0.
Expected R&R of Korean SME Partner	The technical partner will provide the following contributions. In WP1, it will collect requirements and define the architecture. In WP2, it will develop for extracting DPP data from technical diagrams, develop a platform and integrate the development of FhG IOSB's relevant software systems and lead integration testing as well as continuous integration and continuous deployment (CI/CD). In WP3, it will work closely with the pilots and support them in the deployment, use, and evaluation of the AI DPP4Circularity system. In WP4, it will organize activities for the industrial partners and present project results at relevant events and fairs. End-user partner(s) will provide requirements, data of designated products (e.g. product sheets), and access to, or exports from, relevant software systems. They will use the AI DPP4Circularity system to create DPPs for selected products and provide feedback as part of the AI DPP4Circularity system evaluation.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The AI-DPP4Circularity project will equip Korean SMEs with AI-powered technology that makes it feasible to generate, manage, and continuously update DPPs. This automated capability will greatly reduce the technical and administrative burden associated with meeting evolving international product transparency and circularity requirements, including those under the EU's ESPR, which mandates DPPs for many products from 2026 onward. This will help Korean SMEs maintain and expand access to major export markets such as Europe without incurring unnecessary compliance costs or trade barriers. Technologically, the project will deliver interoperable, standards-compliant DPP generation tools that automatically extract, organize, and enrich data from diverse sources, improving data consistency, traceability, and lifecycle transparency across supply chains. These tools will support SMEs in transforming complex datasets into usable digital records that comply with international requirements and enable lifecycle-aware decision-making. In terms of commercialization potential, these capabilities will lower compliance costs and enable new value-added services, such as sustainability reporting, lifecycle analytics, and circular services (e.g., repair support and recycling guidance), strengthening SME competitiveness and opening opportunities in sustainability-driven markets. Standardized DPPs can also serve as a foundation for digital products and services that improve customer engagement and supply chain partnerships. At the broader industrial and economic level, automated DPP adoption will strengthen supply chain transparency and cooperation, enable better resource efficiency, and enhance trust among consumers and business partners by providing reliable, verifiable product information. This supports Korea's transition toward a circular economy, helps Korean firms align with global sustainability practices, and reinforces competitiveness in industries where traceability and environmental performance are increasingly demanded by regulators and markets.

6. References

(1) FA³ST Ecosystem: AI-supported Development and Use of Asset Administration Shells (AAS) - Fraunhofer IOSB

(2) DPP 4.0

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization